

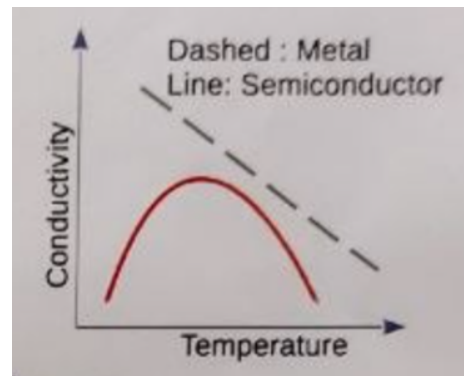
QUIZ 2 (1/9/2026)

Instructions: Do your work neatly and write only in the space provided. TAs and instructor are free NOT to evaluate messy work or answers written out of their place. First, write the solution in your book and only when you are confident to put it on the question.

Maximum Marks [10 points]

1. Justify the difference in qualitative behaviour of the conductivity in the metal (monotonic) and the n-type semiconductor (non-monotonic) as shown. Please write your answers in point form and very neatly, forming a clean justification. Ambiguity in sentences may get you negative marks.

[3 points]



2. Sketch the position of the Fermi level (E_F) in an n-type semiconductor, E_C and E_V (i.e., y-axis is Energy) as a function of temperature (X-axis) on the same graph. Mark the donor energy level (E_D) at $T = 0$ K.

[3 points]

3. A compensated semiconductor has both donor and acceptor impurities. Consider a compensated semiconductor with $N_D > N_A$. Calculate the electron concentration and hole concentration. **[4 points]**